TAGAPINE INJECTION (2 ML AMP)

DESCRIPTION:
TAGAPINE INJECTION (2 ML AMP): A clear, colourless solution.

COMPOSITION:
Each ml contains Cimetidine 100 mg. Each ampoule contains a total of Cimetidine 200 mg per 2 ml.

PHARMACODYNAMICS:
Cimetidine is a histamine H2-receptor antagonist and represents a new class of pharmacological agents. It was the first available agent that blocks the action of histamine at the H2-receptor site and does so by competitive inhibition. Pharmacologically, cimetidine does not exhibit classical anticholinergic effects. Studies have shown that cimetidine inhibits both daytime and nocturnal basal gastric acid secretion. Cimetidine also inhibits gastric acid secretion stimulated by food, histamine, pentagastrin, caffeine and insulin.

PHARMACOKINETICS:
Absorption: Readily absorbed from the gastrointestinal tract. Peak plasma concentration obtained one hour after administration on empty stomach. Second peak may be seen after about 3 hours. Food delays rate of absorption – peak plasma concentration after about 2 hours.
Bioavailability of cimetidine is about 60 to 70% following oral administration. Elimination half-life is around 2 hours and is increased in renal impairment.
Weakly bound about 20% to plasma proteins.
Metabolised partially in the liver to the sulphoxide and to hydroxymethyl cimetidine, but most is excreted unchanged in the urine.
It crosses the placenta barrier and is excreted into breast milk where it is reportedly to be of higher concentration than is in the plasma.

INDICATIONS:
1. The short term treatment of proven duodenal ulcer, gastric ulcer. Maintenance treatment in those patients with recurrence of duodenal ulceration after short term therapy.
2. Maintenance therapy for periods of up to one year to reduce the risk of relapse in patients with documented healing of chronic benign gastric ulcer.
3. Short term treatment (no more than 12 weeks) of persistent gastro-oesophageal reflux disease unresponsive to conservative anti-reflux measures and simple drug therapies such as antacids.
4. Prevention of stress ulcer in critically ill patients at risk of haemorrhage.
5. The treatment of gastrinoma (Zollinger-Ellison syndrome).

RECOMMENDED DOSAGE:
Parenteral: Cimetidine may be administered parenterally in the following ways:
Intermittent i.v. infusion. Intravenous administration may be by intermittent infusion of 200 mg or 300 mg diluted in 100 mL Dextrose Injection (5%) or other compatible intravenous solution infused over 15-20 minutes and repeated every 4-6 hours. In some patients, it may be necessary to increase dosage. When this is necessary, the increase should be made by more frequent administration of the dose but should not exceed 2 g per day.
Continuous i.v. infusion. Continuous intravenous infusion may be used; the average infusion rate should normally not exceed 75 mg/hour during 24 hours.
Intramuscular injection. The usual intramuscular dose is 200mg, which may be repeated at intervals of 4 – 6 hours.
Intravenous injection. In the event that intravenous injection is necessary, 200 mg or 300 mg cimetidine injection should be diluted in 0.9% sodium chloride (or other compatible solution, see (Compatible Solutions; Intravenous Injection) to a total volume of 20 mL and given SLOWLY over a period of not less than 2 minutes. The 200 mg dose may be repeated at 3-6 hourly intervals. The 300 mg dose may be repeated at 4-8 hourly intervals. This method of administration should be avoided in patients with cardiovascular disease and in critically ill patients (see Precautions: Intravenous Injection).
After active bleeding has been stopped, an oral dose regimen may be initiated. The usual dosage is 200 mg three times a day with meals and 400 mg at bedtime. It may be necessary to increase the dosage to 400 mg four times a day.

Compatible Solutions – Intravenous Injection: Although physical and chemical stability of cimetidine injection has been demonstrated for 48 hours at 25°C in the intravenous solutions listed below, it is recommended that to reduce microbial contamination hazards, infusions should be commenced as soon as practicable after preparation of mixtures, and infusion completed within 24 hours and the residue discarded.

Sodium Chloride Injection: 5% and 10% Dextrose.
Lactated Ringer's Solution: 5% Sodium Bicarbonate Injection; 5% Dextrose with 0.2% Sodium Chloride.

Impaired Renal Function: Dosage should be reduced according to creatinine clearance. The following dosages are suggested:

Creatine Clearance (mL/min)	Dosage
0 - 15	200 mg
15 - 30	200 mg three times daily
30 - 50	200 mg four times daily
> 50	Normal dosage

There has been little experience with use of cimetidine constant intravenous infusion in patients with limited renal function. For this reason, administration of cimetidine by constant intravenous infusion is not recommended in patients of this type.
For patients undergoing hemodialysis it is recommended that dialysis be carried out just prior to the next scheduled dosage since some drug will be removed by dialysis. Where circumstances require an increase in dosage, increases should be made by increasing the frequency of administration of 200 mg doses. Cimetidine removal by continuous ambulatory peritoneal dialysis is insignificant and there is no need to adjust conventional renal failure dosage regimen in these patients.

Special Cases: In some instances, e.g. draining gastrocutaneous fistula, control of acid secretion may be necessary. If intravenous administration is required, the dosage schedule should be the same as that recommended above.
Use in Children over 1 year of age: 25-30 mg/kg bodyweight per day in divided doses by mouth or intravenously.

ROUTE OF ADMINISTRATION : For i.m. and slow i.v. infusion

CONTRAINDICATIONS: Known hypersensitivity to Cimetidine.

WARNINGS AND PRECAUTIONS:
Gastric ulcer: Treatment with a histamine H2 antagonist may mask symptoms associated with carcinoma of the stomach and therefore may delay diagnosis of the condition. The potential delay in diagnosis should be borne in mind in patients of middle age or older with new or recently changed dyspeptic symptoms. Accordingly, where gastric ulcer is suspected, the possibility of malignancy should be excluded before therapy with cimetidine is instituted. It is then important to re-endoscope the patient after 8-12 weeks of Cimetidine therapy to check that the ulcer has healed.

In patients with impaired renal function or undergoing haemodialysis: See Dosage and Administration for specific recommendations for use in these patients. It should be borne in mind that some elderly patients may have reduced renal function; if renal function is normal, however, no dosage adjustment is necessary.
In intubated patients receiving mechanical ventilation: Agents that elevate gastric pH may increase the already present risk of nosocomial pneumonia in intubated ICU patients receiving mechanical ventilation.
Intravenous Injection: Rapid intravenous injection of cimetidine should be avoided as there have been rare associations with cardiac arrest and arrhythmias (see *Dosage and Administration: Parenteral*).
Blood pressure and pulse should be monitored during intravenous administration since hypotension and tachycardia/bradycardia have been observed.
Use in children: Clinical experience in children is limited. Therefore, cimetidine therapy cannot be recommended for children unless in the judgement of the physician, anticipated benefits outweigh the potential risks. In limited experience, 25-30 mg/kg bodyweight per day has been administered in divided doses by mouth or intravenously.

INTERACTIONS WITH OTHERS MEDICAMENTS:
Cimetidine, apparently through an effect on certain microsomal enzyme system, have been reported to reduce the hepatic metabolism of warfarin-type anticoagulants, phenytoin, propranolol, chlordiazepoxide, diazepam, lignocaine and theophylline thereby delaying elimination and increasing blood levels of these drugs. Since clinically significant effects have been reported with the warfarin anticoagulants, close monitoring of prothrombin time is recommended and adjustment of the anticoagulant dose may be necessary when cimetidine is administered concomitantly.
Interaction with phenytoin has also been reported to produce adverse clinical effects.
Increased plasma levels of nifedipine have been reported during concomitant cimetidine administration. For concomitant administration of these two drugs, cautious titration of nifedipine is advised.
In order to maintain safe, optimum therapeutic blood levels of the drugs mentioned above and other similarly metabolised drugs, particularly those of low therapeutic index, dosage adjustments may be required when starting or stopping concomitantly administered cimetidine.
Other drugs which may be affected by cimetidine include -blockers, calcium channel blockers, tricyclic anti-depressants, benzodiazepines, chlormethiazole and metformin.

USE IN PREGNANCY AND LACTATION:
Use in Pregnancy: There has been limited experience to date with use of cimetidine in pregnant patients. No significant adverse effects have been reported. Reproductions in rats, mice and rabbits have revealed no evidence of impaired fertility or malformation in the foetus due to cimetidine. However, it has been demonstrated that cimetidine crosses the placental barrier and has been shown to cross the blood brain barrier of neonatal animals. Therefore, cimetidine should only be administered to pregnant patients or women of childbearing potential, when, in the judgement of the physician, the anticipated benefits outweigh the potential risk.
Use in lactation: Cimetidine is excreted in human breast milk, and as a general rule, nursing should not be undertaken while a patient is in on the drug.

SIDE EFFECTS:
In a review of patients in short term clinical trials, cimetidine was found to be well tolerated. Headache 3.1%, diarrhoea 1.8%, tiredness 1.7%, dizziness 1.3%, drowsiness 1.3%, rash 1.2% and constipation 1.0% occurred most frequently. Headache and constipation occurred more frequently in placebo treated patients. Overall the incidence of unwanted effects was comparable between placebo and cimetidine treated groups.
In a review of patients treated with cimetidine in maintenance trials (up to 12 months) the following side effects were reported most commonly. Tiredness 3.5%, headache 2.6%, musculoskeletal pain 2.2%, rash 1.7%, constipation 1.5%, diarrhoea 1.5%, vomiting 1.2%, nausea 1.1%, flatulence 1.1%, depression 1.1%. Headache, diarrhoea, dizziness, nausea and vomiting occurred more commonly in placebo treated patients.
Severe skin rash and reversible alopecia have been reported on occasion. Gynaecomastia and impotence have been reported in some patients receiving high doses. These conditions are usually reversible on discontinuation of cimetidine therapy. The incidence of gynaecomastia and impotence is dependent on dose and duration of treatment. Some increases in serum transaminase and small increases in plasma creatinine have been reported and should be borne in mind when treating patients with renal or hepatic insufficiency. The rises in creatinine have occurred in 11% of patients usually during the first week of treatment and have been non-progressive, returning to pre- treatment values either during therapy or one week after therapy ceased. The significance of these changes is unknown.
Confusional states, reversible within a few days of withdrawing cimetidine have been reported rarely, usually in elderly or ill patients, such as those with renal insufficiency or organic brain syndrome.
Rare cases of hepatitis, interstitial nephritis and pancreatitis which cleared on withdrawal of the drug have been reported.
Rare cases of sinus bradycardia, tachycardia, heart block and anaphylaxis have been reported in patients treated with cimetidine. Cimetidine administration has been rarely associated with the occurrence of leucopenia (including agranulocytosis), thrombocytopenia, pancytopenia and aplastic anaemia. A benefit/risk assessment should be made when concomitant use of cimetidine with drugs known to cause bone marrow depression is contemplated.
Hallucination and depression have been reported very rarely.

SYMPTOMS AND TREATMENT OF OVERDOSE:
Symptoms: There have been reports of severe CNS symptoms such as unresponsiveness following ingestion of between 20g and 40g of cimetidine. There have been deaths in adults who were reported to have ingested over 40g of cimetidine orally as a single dose.
In animal toxicity, CNS depression, hypotension, tachycardia, liver enzyme elevation and renal abnormalities have been observed.
Management: Remove unabsorbed material from the stomach by gastric lavage or Syrup of Ipecac if not more than 4 hours have elapsed since ingestion. Institute supportive therapy for the evolving clinical syndrome. It has been shown in animals that artificial respiration may be of value.

STORAGE CONDITIONS:
Store below 30ºC. Protect from light.
Keep out of reach of children. *Jauhkan daripada kanak-kanak.*

PACK SIZES:
Pack in 10, 30, 50, 100 ampoules of 2 ml / box.

SHELF LIFE:
Please refer to outer package.

PRODUCT REGISTRATION HOLDER AND MANUFACTURER:
DUOPHARMA (M) SDN BHD
Lot 2599, Jalan Seruling 59, Kawasan 3,
Taman Klang Jaya, 41200 Klang, Selangor, MALAYSIA

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